

TITLE: ICU Protocol to consult Ophthalmology for Patients with Suspected Fungal Endophthalmitis

PROTOCOL: Fungal Endophthalmitis

PURPOSE: Endophthalmitis diagnosis in the ICU is often difficult due to its insidious presentation and the common, co-morbid ICU conditions including altered mentation, delirium, and prolonged sedation. Nevertheless, this disease process is highly morbid with the potential of irreversible blindness occurring in affected patients. Historically, the recommendations from the Infectious Disease Society of America (IDSA) and those from the American Academy of Ophthalmology (AAO) concerning consultation of Ophthalmology to evaluate for Endophthalmitis have lacked congruency. This protocol establishes guidelines for the consultation of Ophthalmology for endophthalmitis evaluation in the ICU patient.

APPROVED:



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DEFINITIONS:

- A. AAO – American Academy of Ophthalmology
- B. ISDA – Infectious Disease Society of America
- C. SCCM – Surgical Critical Care Medicine

CORE PROCEDURE:

1. Criteria for Ophthalmology Consult

- a. Visual / Eye Symptoms and one of the below criteria:
 - i. Immunocompromised with suspicion of fungal infection
 - ii. History of fungemia
 - iii. Evidence of fungal septic emboli to other organs
 - iv. Recent Ocular Surgery or Injection
- b. Patient unable to provide reliable history AND with confirmed fungemia:
 - i. Penetrating Ocular Injury
 - ii. Complex Endovascular Source of Infection
 - 1. Vascular Graft
 - 2. Endocarditis
 - 3. Central Line (ONLY if not removed)
 - iii. Persistent Candidemia >72 hours or duration is indeterminable
 - iv. Metastatic Foci of Candidal Infection (i.e. lung, liver, joint)
 - v. History of IV drug use
 - vi. Echinocandin Therapy

2. Timing of Consultation

- a. Emergent (any time, any day) – will be seen within 12 hours
 - i. Presenting with Traumatic Ocular Injury
- b. Non-Urgent (M-F, mornings) - will be seen within 24 – 48 hours
 - i. Fungemia plus:
 - 1. Symptoms (designate in consult and will be seen within 12 hours)
 - OR
 - 2. Exception as listed above
 - ii. History of recent ophthalmologic procedure with symptoms

3. Process for Consultation

- a. Place Order for consult in Epic
 - i. IP Consult to Ophthalmology

- ii. Reason for Consult: Fungal Endophthalmitis Evaluation
- b. Message Ophthalmology via Epic Secure Chat; include:
 - i. MRN (or link message to patient chart)
 - ii. History
 - iii. Signs/symptoms
 - iv. Reason(s) (from above list) that warrant urgent consultation
 - v. If the patient is appropriate for pupil dilation
 - vi. Phone Number at which SCCM can be reached
- c. Non-Emergent Consults should be placed M-F in mornings

REFERENCES:

Breazzano MP, Bond JB 3rd, Bearely S, Kim DH, Donahue SP, Lum F, Olsen TW; American Academy of Ophthalmology. American Academy of Ophthalmology Recommendations on Screening for Endogenous Candida Endophthalmitis. *Ophthalmology*. 2022 Jan;129(1):73-76. doi: 10.1016/j.opthta.2021.07.015. Epub 2021 Jul 19. PMID: 34293405.

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Peter G. Pappas, Carol A. Kauffman, Et. AL., Clinical Practice Guideline for the Management of Candidiasis: 2016 Update by the Infectious Diseases Society of America, *Clinical Infectious Diseases*, Volume 62, Issue 4, 15 February 2016, Pages e1–e50, <https://doi.org/10.1093/cid/civ933>